Hyperspectral Image Classification Using Feature Fusion Hypergraph Convolution Neural Network

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***Abstract*— Convolution neural networks (CNNs) and graph representation learning are two common methods for hyperspectral image (HSI) classification. Recently, graph convolutional neural networks, a combination of CNN and graph representation learning, have shown great potential in the HSI classification problem. However, the existing graph convolution network (GCN)-based methods have many problems, such as overdependence on the adjacency matrix, usage of a single modal feature, and lower accuracy than the mature CNN method. In this article, we propose a feature fusion hypergraph neural network (F^2HNN) for HSI classification. F^2HNN first generates hyperedges from features of different modalities to construct a hypergraph representing multimodal features in HSI. Then, the HSI and the extracted hypergraph are input into the hypergraph convolutional neural network for learning. In addition, we propose three feature fusion strategies. The first strategy is the most basic spatial and spectral feature fusion. The second strategy fuses the spectral features extracted by a pretrained multilayer perceptron (MLP) with the spatial features to reduce the redundant information of the original spectral features. The third strategy uses the fusion of CNN features, spectral features, and spatial features to explore the capabilities of F^2HNN. Sufficient experiments on four datasets have proved the effectiveness of F^2HNN.**

***Index Terms*— Deep learning, feature fusion, graph convolution networks (GCNs), hypergraph learning, hyperspectral image (HSI) classification.**

I. Introduction

REMOTE sensing images have received extensive attention in recent decades. In particular, due to the advancement of spectral imaging technology, hyperspectral images (HSIs) have been widely used in food safety detection, medical aid diagnosis, and land-cover classification [1]. HSI refers to image data with continuous spectral resolution and narrowband obtained by using hyperspectral imaging technology [2]. Depart from traditional panchromatic images or multispectral images, HSI has a more satisfying performance and greater research values in the classic remote sensing task of land-cover classification due to the extra massive spectral information. The enormous abundance of data also brings redundancy of information. How to extract effective information from HSI and discover the potential structural features of the data has become widely discussed by researchers in the task of HSI classification.

Identical to most computer vision research, methods in the HSI classification field can be roughly divided into deep learning methods and nondeep learning methods. Before the great success of deep learning, most of the methods used the non-end-to-end strategy, i.e., first extracting features and then selecting the classifier for classification. In the feature extraction strategies, the researchers' work is centered on two aspects. The first one is to reduce the dimensionality of HSI to accommodate the Hughes phenomenon [3], such as using manual feature selection [4] and dictionary-based sparse representation [5], or simply applying principal component analysis (PCA) [6]. The other one is to adopt effective strategies to combine spatial and spectral features [7]–[10]. The manifold learning method in the graph embedding framework has once become a quite prevalent feature extraction scheme [11], [12]. This type of method utilizes graph Laplacian of spectral features to reduce dimensionality [13] and can fuse spatial–spectral features by establishing the adjacency matrix through the context information and the spectral information.

Under the influence of the rapid development of deep learning in recent years, many deep learning methods have been proposed and applied to HSI classification problems [3]. Hu *et al.* [14] first use convolution neural networks (CNNs) to directly classify HSI in the spectral domain. Hang *et al.* [15] proposed a method regarding the spectral signature as a sequence and using recurrent neural network (RNN) with several convolution layers to extract discriminative features. The most significant difference between deep-learning-based methods and traditional methods is that deep learning adopts an end-to-end framework, using multilayer network structure and learning methods to obtain the best feature expression, while traditional methods are based on handcrafted features.

Among the deep learning methods, the CNN-based method is the most widely used and has the best performance. Lee *et al.* [16] designed a CNN model with multiscale convolutional filter bank and residual learning for HSI classification. Chen *et al.* [17] adopted 3-D convolution to extract spectral–spatial features and perform end-to-end training. In spite of the fact that the classification accuracy of HSI has been visibly improved, there are still some shortcomings in

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the development of methods based on CNN. For instance, the convolution kernels have limited receptive fields; thus, it is unable to focus on long-distance dependencies. Moreover, most of the existing CNN-based methods adopt a strategy of cutting the entire HSI into small patches containing dozens of pixels, which loses a lot of nonlocal information.

Due to the successful application of graph embedding structure and CNN in the field of HSI classification, a novel graph convolution network (GCN) [18] has also been tried for HSI classification. GCN is a combination and evolution of graph embedding learning and deep learning, aiming to find a more efficient feature expression method for graph data. There are usually two classification tasks that GCN focuses on: one is the classification of the entire graph data, and the other is the classification of each node in the graph data. The GCNs used for these two different tasks are not exactly the same. For the HSI classification problem, it can be transformed into a node classification problem that GCN can handle [19]. Most of the current GCN methods regard each pixel as a node and the spectral information of each pixel as the feature of the node, thereby utilizing a specific distance metric method to construct graph data from HSI [19], [20]. Hong *et al.* [21] constructed an end-to-end network cascading CNN and GCN, and trained GCN in minibatches. In [22], superpixel segmentation and multistep dynamic map construction were used for GCN training. The above methods prove that there exist potential graph structures in HSI.

Nevertheless, the current GCN-based methods still have some defects. For instance, existing methods rely heavily on the construction of an adjacency matrix, which represents the topological structure of graph data. However, the adjacency matrix is basically built on prior knowledge and cannot be updated through the learning process. Moreover, modeling with a simple graph structure, in which edges can only represent pairwise relationships, cannot discover potentially complex structures in the data, such as hypergraph structures.

In order to solve these problems, we adopt the recently proposed hypergraph neural network (HGNN) [23] improved from GCN and specifically ameliorate the HGNN network to a feature fusion hypergraph neural network (F^2HNN) for HSI data. As a representation learning method, HGNN adopts a hypergraph structure for modeling and generalizes graph convolution to a hypergraph structure. The main difference between the hypergraph structure and the simple graph structure lies in the definition of the edge. The edge in the simple graph structure contains two vertices; in other words, the degree of the edge is forced to 2. In contrast, the hyperedge in the hypergraph can contain any number of nodes and is, therefore, degree-free, as shown in Fig. 1. This difference makes HGNN have more powerful representation capabilities. In addition, due to the existence of the hyperedge weight matrix, it does not rely too much on the construction of the original adjacency matrix. Furthermore, F^2HNN uses a feature fusion strategy based on HGNN. By using different features to generate hyperedges (such as spectral–spatial features) and setting the weights of the hyperedges to trainable parameters, F^2HNN updates the weights of the hyperedges in iterations to reach the optimum.

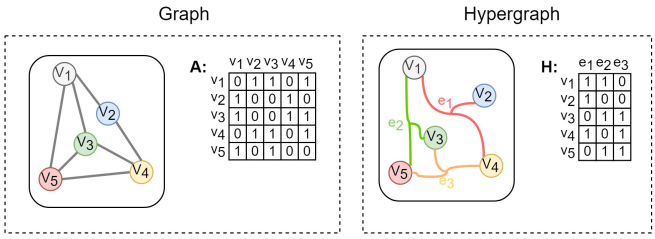


Fig. 1. Comparison of graph and hypergraph structures. The graph structure is usually represented by adjacency matrix **A**, while the hypergraph structure is represented by incidence matrix **H**.

The contribution of this article can be summarized as follows.

1) We propose a novel improved HGNN network named F^2HNN for HSI classification. F^2HNN effectively fuses multimodal features and automatically updates the hyperedge weights, which solves the difficulties encountered by the previous GCN-based hyperspectral classification methods.
2) We design three different feature fusion strategies and conduct experiments on multiple datasets to verify the effectiveness of F^2HNN. Furthermore, we explore the relationship between GCN and CNN by analyzing the results of different strategies on different datasets.
3) Compared with the previous GCN-based network, the proposed F^2HNN is a concise but efficacious end-to-end network. When simply fusing the spectral–spatial features, it can achieve state-of-the-art accuracy.

II. Related Works

In this section, we first review the development of hypergraph learning and its successful application in the field of computer vision. Then, we introduce some researches on HSI classification using graph neural networks (GNNs), summarize the characteristics of representative methods, and infer the possible development directions.

A. Hypergraph Learning

A hypergraph is a generalized graph structure that contains a set of vertices and hyperedges. The edge in the simple graph structure only connects two vertices with the degree of 2, while hyperedges in the hypergraph structure are able to connect multiple vertices and are degree-free. Compared with simple graphs, hypergraphs express more complex relationships than pairwise relationships, which can better model many practical data, such as social networks and citation networks.

Hypergraph learning is generalized from graph learning and was first proposed in [24]. The purpose of hypergraph learning is to analyze data by learning the mode of information transfer between hypergraph structures. The typical tasks of hypergraph learning include node classification [25], link prediction [26], community detection [27], and so on. Due to the fact that hypergraph learning can establish high-order correlation models between data, it has been widely used in many fields, including computer vision. For instance, Wang *et al.* [28] introduced hypergraph learning into image retrieval. In the task

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of image classification, Yu *et al.* [29] generated hyperedges by linking the image with its nearest neighbors and obtained the image labels and hyperedge effects at the same time through adaptive hypergraph learning. In addition, hypergraph learning has also been successfully applied to computer vision tasks, such as person rerecognition [30], 3-D object classification [31], and video segmentation [32].

After the deep learning method revolutionized the area of machine learning, a series of studies that combined deep learning and graph representation learning appeared. GNN was first proposed in [33], and then, various models have been gradually developed, such as graph convolution network [34], graph attention network [35], and GraphSAGE (SAmple and aggreGatE) [36]. As a promotion of graph learning, hypergraph learning is also affected by deep learning. Feng *et al.* [23] first proposed an HGNN and introduced a spectral convolution operator on a static hypergraph. In [37], a dynamic hyperGCN was proposed. Bai *et al.* [38] embedded the two trainable operators of hypergraph convolution and hypergraph attention in GNN to extract the nonpairwise relationships and proved the effectiveness of their method on the semisupervised node classification task.

B. Graph Neural Networks for HSI Classification

It is a conventional method to treat HSI as graph data and perform subsequent processing. Previous researchers used manifold learning to extract non-Euclidean relationships in HSI data. Sufficient graph learning methods have been tried for HSI classification. Camps-Valls *et al.* [39] adopted graph-based composite-kernel semisupervised learning for classification. Gao *et al.* [40] proposed a bilayer graph representation learning method. In recent years, the GNN method has become a research hotspot of HSI classification. Qin *et al.* [19] applied GCN to hyperspectral classification and improved the original GCN to facilitate the fusion of spatial and spectral features. The ideas of [22] and [41] are similar. They both use superpixel segmentation first and then input the segmented superpixels as nodes into the GNN for node classification. Hong *et al.* [21] explored the similarities and differences between GCNs and CNNs in the HSI classification task and proposed a classification framework by combining CNN and GCN. He *et al.* [42] adopted the common strategy of the CNN-based method, divided the HSI into a set of patches, and inputted the patches into the CNN network to extract features. Then, the graph data were generated from the patch with extracted features and inputted into GCN for classification.

III. Proposed Method

The complete framework of our proposed F^{2}HNN is shown in Fig. 2. We first design a strategy to extract multimodal features from HSI and then generate hypergraphs for these features. After having the hypergraph structure, HSI and the generated hypergraph structure are sent to HGNN for training. In this section, F^{2}HNN will be introduced in detail from three aspects: HGNN, hypergraph generation, and feature fusion strategy.

A. HGNN

1) Definition of Hypergraph: A simple undirected graph can be represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ by the set of vertices $\mathcal{V}$ and edges $\mathcal{E}$. Unlike the simple graph structure, the hyperedge in the hypergraph is degree-free, which means that the hyperedge can connect more than two vertices. In addition, each hyperedge e also has a hyperedge confidence parameter $w(e)$ (usually we use weighted hypergraphs in the hypergraph learning method [24]). Therefore, a hypergraph can be defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$, where the diagonal matrix $\mathbf{W}$ is the hyperedge weight matrix. Different from GCN using adjacency matrix $\mathbf{A}$ to represent graph data, HGNN utilizes incidence matrix $\mathbf{H}$ to denote hypergraph data. Both the rows and columns of $\mathbf{A}$ (size $|\mathcal{V}| \times |\mathcal{V}|$) depict vertices, while the rows and columns of $\mathbf{H}$ (size $|\mathcal{V}| \times |\mathcal{E}|$) represent nodes and edges, respectively, as shown in the following:

$$h(v, e) = \begin{cases} 1, & \text{if } v \in e \\ 0, & \text{if } v \notin e. \end{cases} \tag{1}$$

Given an H matrix, the hypergraph Laplacian matrix can be calculated as

$$\mathbf{L} = \mathbf{I} - \mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^\top\mathbf{D}_v^{-1/2} \tag{2}$$

where $\mathbf{D}_v$ and $\mathbf{D}_e$ denote the diagonal matrix of vertex degrees and edge degrees, with each vertex degree defined as $d(v) = \sum_{e\in\mathcal{E}} \omega(e)h(v, e)$ and each edge degree defined as $\delta(e) = \sum_{v\in\mathcal{V}} h(v, e)$. The role of $\mathbf{D}_v$ and $\mathbf{D}_e$ can be simply summarized as normalizing incidence matrix $\mathbf{H}$.

2) Convolution Operation on Hypergraph: Graph convolution is based on spectral graph theory (SGT) [43], [44]. In brief, SGT adopts the eigenvalues and eigenvectors of the graph Laplacian matrix to study the properties of the graph. How the graph convolution operation is calculated has been introduced in detail in [21], and the convolution of the hypergraph is derived from it. Given a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$, the Fourier transform of a signal (vertex) $\mathbf{x} = (x_1, \ldots, x_n)$ is defined as

$$\hat{\mathbf{x}} = \mathbf{\Phi}^\top\mathbf{x} \tag{3}$$

where $\mathbf{\Phi}$ can be calculated by diagonalizing the positive semidefinite matrix $\mathbf{L}$ as

$$\mathbf{L} = \mathbf{\Phi}\mathbf{\Lambda}\mathbf{\Phi}^\top \tag{4}$$

where $\mathbf{\Phi} = \text{diag}(\phi_1, \ldots, \phi_n)$ contains the orthonormal eigen vectors, and $\mathbf{\Lambda} = \text{diag}(\lambda_\mathbf{1}, \ldots, \lambda_\mathbf{n})$ is a diagonal matrix composed of eigenvalues. Then, the hypergraph convolution operation of the signal $\mathbf{x}$ and filter $\mathbf{g}$ can be denoted as

$$\mathbf{g} \star \mathbf{x} = \mathbf{\Phi}\left(\left(\mathbf{\Phi}^\top\mathbf{g}\right) \odot \left(\mathbf{\Phi}^\top\mathbf{x}\right)\right) = \mathbf{\Phi}g(\mathbf{\Lambda})\mathbf{\Phi}^\top\mathbf{x} \tag{5}$$

where $g(\mathbf{\Lambda}) = \text{diag}(\mathbf{g}(\lambda_1), \ldots, \mathbf{g}(\lambda_\mathbf{n}))$ is a function of the Fourier coefficients (can also be seen as the convolution kernel in graph convolution) and $\odot$ denotes the Hadamard product. In order to reduce the computational complexity of $\mathcal{O}(n^2)$ caused by calculating the eigenvalues and eigenvectors of $\mathbf{L}$,

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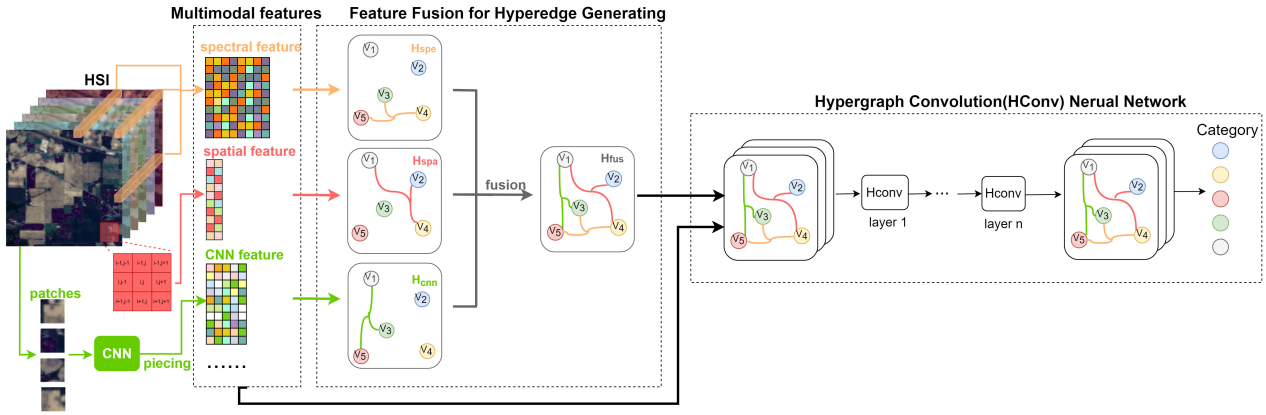


Fig. 2. Overview of our proposed F^{2}HNN. Taking the third fusion strategy as an example, this figure first shows the fusion of spectral features, spatial coordinate features, and CNN features. Next, it shows the training process of the HGNN and the acquisition of classification results.

the Chebyshev polynomial [18] can be utilized to fit the convolution kernel $g(\mathbf{\Lambda})$ as

$$g(\mathbf{\Lambda}) = \sum_{k=0}^{K-1} \beta_k T_k\left(\tilde{\mathbf{\Lambda}}\right) \tag{6}$$

where $\tilde{\mathbf{\Lambda}}$ is the rescaled $\mathbf{\Lambda}$ to ensure that the input of the Chebyshev polynomial is between $[1, -1]$, and T_k is the K-order Chebyshev polynomial that can be computed by

$$T_k(x) = \cos(k \cdot \arccos(x)). \tag{7}$$

Substituting (6) and (7) into (5), we can get

$$\mathbf{g} \star \mathbf{x} \approx \sum_{k=0}^{K} \theta_k T_k\left(\tilde{\mathbf{L}}\right)\mathbf{x} \tag{8}$$

where $\tilde{\mathbf{L}}$ is the rescaled Laplacian $\tilde{\mathbf{L}} = (2/\lambda_{\max})\mathbf{L} - \mathbf{I}$. After reducing the computational complexity, we further set $K = 1$ and $\lambda_{\max} \approx 2$ suggested by Feng *et al.* [23] and Kipf and Welling [34]. Thus, the convolution operation on hypergraph can be further simplified as

$$\mathbf{g} \star \mathbf{x} \approx \theta_0\mathbf{x} - \theta_1\mathbf{D}^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_{\mathbf{e}}^{-1}\mathbf{H}^{\top}\mathbf{D}_{\mathbf{v}}^{-1/2}\mathbf{x} \tag{9}$$

where θ_0 and θ_1 can be integrated by one parameter θ to avoid the overfitting, which is defined as

$$\begin{cases} \theta_1 = -\dfrac{1}{2}\theta \\ \theta_0 = \dfrac{1}{2}\theta\mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{D}_e^{-1}\mathbf{H}^{\top}\mathbf{D}_v^{-1/2}. \end{cases} \tag{10}$$

Then, the convolution operation on hypergraph is further derived as

$$\begin{aligned} \mathbf{g} \star \mathbf{x} &\approx \frac{1}{2}\theta\mathbf{D}_v^{-1/2}\mathbf{H}(\mathbf{W} + \mathbf{I})\mathbf{D}_e^{-1}\mathbf{H}^{\top}\mathbf{D}_v^{-1/2}\mathbf{x} \\ &\approx \theta\mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^{\top}\mathbf{D}_v^{-1/2}\mathbf{x} \end{aligned} \tag{11}$$

where $\mathbf{W}$ represents the weight matrix of the hyperedges, which is usually calculated in advance or directly initialized as an identity matrix.

Given a hypergraph data $\mathbf{X}^{n\times c_1}$ with n vertices and c_1 feature channels, the convolution operation on hypergraph can be formulated by

$$\mathbf{Y} = \mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^{\top}\mathbf{D}_v^{-1/2}\mathbf{X}\mathbf{\Theta} \tag{12}$$

where $\mathbf{W} = \mathrm{diag}(w_1, w_2, \ldots, w_n)$ and $\mathbf{\Theta}^{c_1\times c_2}$ are the trainable parameters. $\mathbf{Y}^{n\times c_2}$ is the output after the convolution operation.

3) Hypergraph Convolution Neural Networks: The complete hypergraph convolutional layer is obtained by the hypergraph convolution operation described above plus the nonlinear activation function, which can be formulated as

$$\mathbf{X}^{(l+1)} = \sigma\left(\mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^{\top}\mathbf{D}_v^{-1/2}\mathbf{X}^{(l)}\mathbf{\Theta}^{(l)}\right) \tag{13}$$

where $\mathbf{X}^{(l+1)}$ is the output of the lth layer and σ is the RELU function used for nonlinear activation.

B. Feature Fusion for Hyperedge Generating

We use the incidence matrix $\mathbf{H}$ to represent the topological structure of the hypergraph. For HSIs, the incidence matrix $\mathbf{H}$ cannot be obtained directly and needs to be generated from the image. GCN-based methods usually use single-modal features for graph construction and learning. This is because they utilize adjacency matrix $\mathbf{A}$ as input, which limits the number of edges. However, the fusion of multimodal features, such as spatial features and spectral features [45]–[47], proved to be very effective in HSI classification tasks. Our proposed F^{2}HNN fuses multimodal features to generate hyperedges and, thereby, needs to obtain the incidence matrix $\mathbf{H}$.

Given HSI data $\mathbf{Z}^{\mathrm{rows}\times\mathrm{cols}\times c}$, we treat each pixel as a vertice of the hypergraph and flatten the image to $\mathbf{X}^{n\times c}$, where $n = \mathrm{rows} \times \mathrm{cols} = |\mathcal{V}|$ represents the number of hypergraph vertices and c represents spectral dimensions. Then, we use different feature extractors to extract features of different modalities $\mathbf{X}_i^{n\times c_i}$ from the original $\mathbf{X}^{n\times c}$.

For each vertex $v \in \mathcal{V}$ in $\mathbf{X}$, we select its k nearest neighbors to generate a hyperedge $e \in \mathcal{E}$; thus, we obtain an incidence matrix $\mathbf{H}^{|\mathcal{V}|\times|\mathcal{E}|}(|\mathcal{V}| = |\mathcal{E}| = n)$

$$h(i, j) = \begin{cases} e^{-\sigma\|\mathbf{x}_i - \mathbf{x}_j\|^2/\mathbf{mean}}, & \text{if } \mathbf{x}_i \in \mathcal{N}_k(\mathbf{x}_j) \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where **mean** represents the average of the Euclidean distance between all vertices $v \in \mathcal{V}$, and σ is an adjustable hyperparameter. The purpose of using **mean** in this equation is to normalize the distances of multimodal features and facilitate the adjustment of hyperparameters.

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Given multimodal features $[\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m]$, for each $\mathbf{X}_i$, we use (14) to calculate $\mathbf{H}_i$, and then, we get $[\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_m]$. The way that we fuse the hyperedge obtained by multimodal features is to concatenate these incidence matrices into $\mathbf{H}_{\text{fuse}} = [\mathbf{H}_1\ \mathbf{H}_2, \ldots, \mathbf{H}_m]$. Then, the hypergraph convolution of (12) becomes

$$\begin{aligned}\mathbf{Y} &= \mathbf{D}_v^{-1/2}\mathbf{H}_{\text{fuse}}\mathbf{W}_{\text{fuse}}\mathbf{D}_v^{-1/2}\mathbf{H}_{\text{fuse}}^T\mathbf{D}_e^{-1}\mathbf{X}\boldsymbol{\Theta} \\ &= \mathbf{D}_v^{-1/2}\begin{bmatrix}\mathbf{H}_1 & \cdots & \mathbf{H}_m\end{bmatrix}\begin{bmatrix}\mathbf{W}_1 & & \\ & \cdots & \\ & & \mathbf{W}_m\end{bmatrix}\mathbf{D}_e^{-1}\begin{bmatrix}\mathbf{H}_1^T \\ \cdots \\ \mathbf{H}_m^T\end{bmatrix} \\ &\quad \mathbf{D}_v^{-1/2}\mathbf{X}\boldsymbol{\Theta}. \end{aligned} \tag{15}$$

If not considering the normalization matrices $\mathbf{D}_v$ and $\mathbf{D}_e$, (15) can be transformed to

$$\begin{aligned}\mathbf{Y}' &= \mathbf{H}_{\text{fuse}}\mathbf{W}_{\text{fuse}}\mathbf{H}_{\text{fuse}}^T\mathbf{X}\boldsymbol{\Theta} \\ &= \begin{bmatrix}\mathbf{H}_1 & \cdots & \mathbf{H}_2\end{bmatrix}\begin{bmatrix}\mathbf{W}_1 & & \\ & \cdots & \\ & & \mathbf{W}_m\end{bmatrix}\begin{bmatrix}\mathbf{H}_1 \\ \cdots \\ \mathbf{H}_m\end{bmatrix}\mathbf{X}\boldsymbol{\Theta} \\ &= \left(\mathbf{H}_1\mathbf{W}_1\mathbf{H}_1^T + \cdots + \mathbf{H}_m\mathbf{W}_m\mathbf{H}_m^T\right)\mathbf{X}\boldsymbol{\Theta}. \end{aligned} \tag{16}$$

Equation 16 shows that using $\mathbf{H}_{\text{fuse}}$ for training in hypergraph convolution is equivalent to performing collaborative training on each subgraph $\mathbf{H}_i$ represented by each feature $\mathbf{X}_i$.

Algorithm 1 Pseudocode of the Fusion Strategy of "Original Spectral Feature + Spatial Feature"

input Original HSI $\mathbf{X}_{ori}$, number of epochs $\mathcal{E}$, number of layers $\mathcal{L}$, number of neighbors k
Generate $\mathbf{X}_{spe}$ by flatting $\mathbf{X}_{ori}$
Generate $\mathbf{X}_{spa}$ according to Equation 17
$\mathbf{X} = [\mathbf{X}_{spe}\ \ \mathbf{X}_{spa}]$
Construct $\mathbf{H}_{spe}$ and $\mathbf{H}_{spa}$ according to Equation 14 and k
$\mathbf{H} = [\mathbf{H}_{spe}\ \ \mathbf{H}_{spa}]$
Calculate $\mathbf{D}_v$ and $\mathbf{D}_e$ from $\mathbf{H}$
Initialize $\boldsymbol{\Theta}$ and $\mathbf{W}$
for $i = 1$ to $\mathcal{E}$ **do**
  for $l = 1$ to $\mathcal{L}$ **do**
    $\mathbf{X}^l = \sigma(\mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^\top\mathbf{D}_v^{-1/2}\mathbf{X}^{l-1}\boldsymbol{\Theta}^{l-1})$
  end for
  $\mathbf{X}_{pre} = SOFTMAX(\mathbf{X}^{\mathcal{L}})$
  Calculate loss and update parameters $\boldsymbol{\Theta}$ and $\mathbf{W}$ through gradient backpropagation
end for
output The predicted label of each pixel in HSI

C. Different Fusion Strategies

After obtaining a network framework suitable for multimodal feature fusion and training, how to obtain and select multimodal features has become a matter of concern. In this section, we separately introduce three representative feature fusion strategies used in this article, including the original unprocessed HSI data features and the features extracted by the pretrained feature extractor.

Algorithm 2 Pseudocode of the Fusion Strategy of "Spectral Feature Extracted by MLP + Spatial Feature"

input Original HSI $\mathbf{X}_{ori}$, number of epochs $\mathcal{E}$, number of layers $\mathcal{L}$, number of neighbors k
Generate $\mathbf{X}_{spe}$ by flatting $\mathbf{X}_{ori}$
Use $\mathbf{X}_{spe}$ as input to pretrain a MLP, and obtain $\mathbf{X}_{mlp} = MLP(\mathbf{X}_{spe})$
Generate $\mathbf{X}_{spa}$ according to Equation 17
$\mathbf{X} = [\mathbf{X}_{mlp}\ \ \mathbf{X}_{spa}]$
Construct $\mathbf{H}_{mlp}$ and $\mathbf{H}_{spa}$ according to Equation 14 and k
$\mathbf{H} = [\mathbf{H}_{mlp}\ \ \mathbf{H}_{spa}]$
Calculate $\mathbf{D}_v$ and $\mathbf{D}_e$ from $\mathbf{H}$
Initialize $\boldsymbol{\Theta}$ and $\mathbf{W}$
for $i = 1$ to $\mathcal{E}$ **do**
  for $l = 1$ to $\mathcal{L}$ **do**
    $\mathbf{X}^l = \sigma(\mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^\top\mathbf{D}_v^{-1/2}\mathbf{X}^{l-1}\boldsymbol{\Theta}^{l-1})$
  end for
  $\mathbf{X}_{pre} = SOFTMAX(\mathbf{X}^{\mathcal{L}})$
  Calculate loss and update parameters $\boldsymbol{\Theta}$ and $\mathbf{W}$ through gradient backpropagation
end for
output The predicted label of each pixel in HSI

1) Original Spectral Feature + Spatial Feature: Spectral–spatial feature fusion is the most common and effective feature fusion strategy in the HSI classification task. Consider the most general case; we first use the original unprocessed spectral feature and the spatial feature. The spatial feature $\mathbf{X}_{\text{spatial}}^{n\times 2}$ is obtained through pixel coordinates

$$\mathbf{X}_{\text{spatial}}[i] = [x(i), y(i)] \tag{17}$$

where $x(i)$ and $y(i)$ represent the horizontal and vertical coordinates of pixel i, respectively. Algorithm 1 summarizes the process of the proposed strategy.

2) Spectral Feature Extracted by Multilayer Perceptron + Spatial Feature: In the task of hyperspectral classification, preprocessing of spectral features has proved to be necessary. Therefore, in the second strategy, we use a pretrained multilayer perceptron (MLP) as a feature extractor to preprocess the spectral features and then fuse them with the spatial features obtained in (17). The processing flow of this strategy is shown in Algorithm 2.

3) Spectral Feature + Spatial Feature + CNN Feature: The CNN method in the HSI classification task divides a single HSI into different patches containing local information and inputs them into the network for training. This type of method effectively extracts the local information contained in the HSI patches. However, due to the existence of patches, the CNN method ignores nonlocal information. Unlike the CNN method, the graph constructed with spectral information discards the local information contained in the Euclidean data and focuses on long-distance dependencies. Our third fusion strategy considers both the fusion of spectral–spatial features and the fusion of local and nonlocal information, and fuses the three features of the spectral feature, the coordinate-encoded spatial feature, and the CNN feature extracted by a pretrained model. Algorithm 3 indicates the process of the third strategy.

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Algorithm 3 Pseudocode of the Fusion Strategy of "Spectral Feature + Spatial Feature + CNN Feature"

input Original HSI $\mathbf{X}_{ori}$, number of epochs $\mathcal{E}$, number of layers $\mathcal{L}$, number of neighbors k

Generate $\mathbf{X}_{spe}$ by flatting $\mathbf{X}_{ori}$
Use a set of patches $\mathcal{P}$ generated from $\mathbf{X}_{spe}$ as input to pretrain a CNN model
for each pixel j in HSI **do**
  Generate a 7×7 patch $\mathcal{P}_j$ with j as the center
  $x_j = CNN(\mathcal{P}_j)$
  Generate $\mathbf{X}_{cnn}$ by piecing each x_j
end for
Generate $\mathbf{X}_{spa}$ according to Equation 17
$\mathbf{X} = [\mathbf{X}_{spe} \quad \mathbf{X}_{spa} \quad \mathbf{X}_{cnn}]$
Construct $\mathbf{H}_{spe}$, $\mathbf{X}_{spa}$ and $\mathbf{H}_{cnn}$ according to Equation 14 and k
$\mathbf{H} = [\mathbf{H}_{spe} \quad \mathbf{H}_{spa} \quad \mathbf{X}_{cnn}]$
Calculate $\mathbf{D}_v$ and $\mathbf{D}_e$ from $\mathbf{H}$
Initialize $\mathbf{\Theta}$ and $\mathbf{W}$
for $i = 1$ to $\mathcal{E}$ **do**
  for $l = 1$ to $\mathcal{L}$ **do**
    $\mathbf{X}^l = \sigma(\mathbf{D}_v^{-1/2}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^\top\mathbf{D}_v^{-1/2}\mathbf{X}^{l-1}\mathbf{\Theta}^{l-1})$
  end for
  $\mathbf{X}_{pre} = SOFTMAX(\mathbf{X}^{\mathcal{L}})$
  Calculate loss and update parameters $\mathbf{\Theta}$ and $\mathbf{W}$ through gradient backpropagation
end for

output The predicted label of each pixel in HSI

IV. Experiment

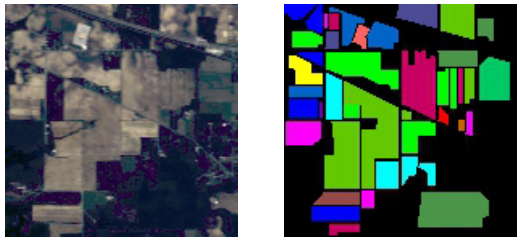
A. Datasets

We select four representative datasets and perform comprehensive experiments to verify the effectiveness of our proposed method.

1) Indian Pines: The Indian Pines dataset was photographed in Northwestern Indiana by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The size of the data is 145 × 145 with 20 m × 20 m spatial resolution. The image contains 220 spectral bands covering 400–2500 nm, of which 20 water absorption and noisy bands are removed. A total of 16 classes of land covers are labeled for classification. Table I details the total number of samples for each category and the number of samples used for training. Fig. 3 shows the pseudocolor image and the ground-truth map of Indian Pines.

2) Kennedy Space Center: The Kennedy Space Center (KSC) dataset was collected in Florida by the AVIRIS sensor. It contains 224 spectral bands sampled from 400 to 2500 nm and 614 × 512 pixels with a spatial resolution of 18 m × 18 m. The selected 176 bands and 13 labeled classes of land covers are used for classification. Table III lists the total number of samples for each category and the number of samples used for training. The pseudocolor image and the ground-truth map of KSC are shown in Fig. 4. Training and testing samples for KSC are listed in Table II.

3) Botswana: The Botswana dataset was acquired by NASA's EO-1 satellite over the Okavango Delta, Botswana.

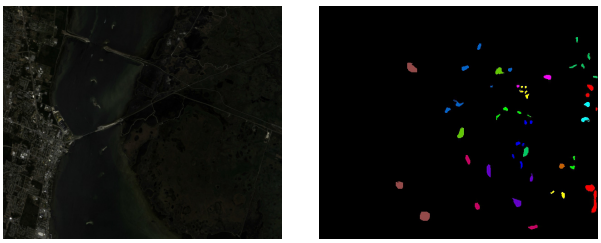


(a) (b)

Fig. 3. Visualization of the India Pines dataset. (a) Pseudocolor map. (b) Ground-truth map.

TABLE I
Number of Training Samples and Test Samples for Each Class of Land Cover in Indian Pines

Class No.	Class Color	Class Name	Training	Testing
1		Alfalfa	15	31
2		Corn Notill	50	1378
3		Corn Mintill	50	780
4		Corn	50	187
5		Grass Pasture	50	433
6		Grass Trees	50	680
7		Grass Pasture Mowed	15	13
8		Hay Windrowed	50	428
9		Oats	15	5
10		Soybean Notill	50	922
11		Soybean Mintill	50	2405
12		Soybean Clean	50	543
13		Wheat	50	155
14		Woods	50	1215
15		Buildings Grass Trees Drives	50	336
16		Stone Steel Towers	50	43
		Total	695	9554



(a) (b)

Fig. 4. Visualization of the KSC dataset. (a) Pseudocolor map. (b) Ground-truth map.

The sensor on EO-1 obtained 1476 × 256 data with a 30-m pixel resolution and 242 spectral bands covering a range of 400–2500 nm. It includes 14 land-cover classes. Since the atmospheric and water absorption bands are invalid, 145 out of the 242 bands are selected. Training and testing samples for Botswana are shown in Table III. Fig. 5 shows the pseudocolor image and the ground-truth map of Botswana.

4) Pavia University: The Pavia University dataset is acquired by ROSIS sensors. After removing noisy bands, it has a total of 103 effective bands, covering the electromagnetic spectrum from 400 to 860 nm. The dataset contains nine land-cover classes, and the size is 610 × 340. The spatial resolution of the dataset is 1.3 m × 1.3 m. Table IV lists the number of training samples. Fig. 6 shows the pseudocolor image and the ground-truth map of Pavia University.

B. Experimental Settings

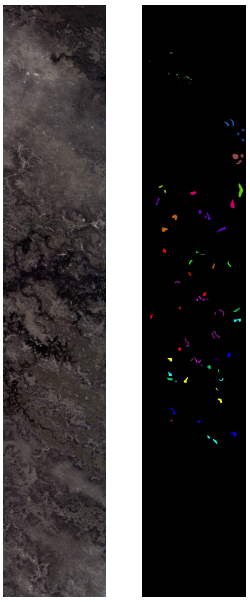
In this section, we first introduce the detailed structure of our proposed F^{2}HNN and the parameter settings of the

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TABLE II

NUMBER OF TRAINING SAMPLES AND TEST SAMPLES FOR EACH CLASS OF LAND COVER IN KSC

Class No.	Class Color	Class Name	Training	Testing
1		Srub	30	728
2		Willow swamp	30	220
3		CP hammock	30	232
4		Slash pine	30	228
5		Oak/Broadleaf	30	146
6		Hardwood	30	207
7		Swamp	30	96
8		Graminoid	30	393
9		Spartina marsh	30	469
10		Cattail marsh	30	365
11		Salt marsh	30	378
12		Mud flats	30	454
13		Water	30	836
		Total	390	4752



(a) (b)

Fig. 5. Visualization of the Botswana dataset. (a) Pseudocolor map. (b) Ground-truth map.

experiments. Then, we show all the baseline models used for comparison and the settings of their parameters.

1) Detailed Supplement of Proposed F²HNN: The three feature fusion strategies used in the experiments require two pretrained models, i.e., the MLP model adopted in the second strategy and the CNN model used in the third strategy. We design a three-layer MLP and a four-layer 3-D CNN as the pretrained model. The detailed parameters are shown in Table V. The training samples of the pretrained model are consistent with the training samples in the subsequent F²HNN. It is worth noting that, for the Pavia University dataset with large image size, we averagely cut it into four parts for classification according to the number of labeled samples included.

The network is implemented using the PyTorch framework, the experiment is set to full gradient descent, and Adam [48] is used for optimization with a weight decay set to 0.0005. The number of total epochs is chosen as 200, and the learning rate is set to be dynamically adjustable. The optimal initial learning rate of 0.01 is chosen from {0.0001, 0.001, 0.01, 0.1, 1}, and the learning rate drops to one-half of the previous one after 30 epochs.

TABLE III

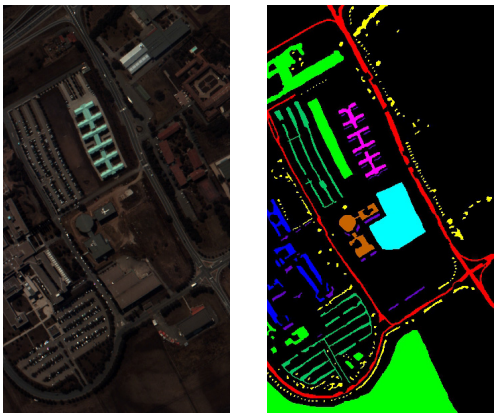
NUMBER OF TRAINING SAMPLES AND TEST SAMPLES FOR EACH CLASS OF LAND COVER IN BOTSWANA

Class No.	Class Color	Class Name	Training	Testing
1		Water	10	270
2		Hippo grass	10	101
3		Floodplain grasses 1	5	251
4		Floodplain grasses 2	5	215
5		Reeds	5	269
6		Riparian	5	269
7		Firescar	5	259
8		Island interior	5	203
9		Acacia woodlands	5	314
10		Acacia shrublands	5	248
11		Acacia grasslands	5	305
12		Short mopane	5	181
13		Mixed mopane	5	268
14		Exposed soils	5	95
		Total	140	3248

TABLE IV

NUMBER OF TRAINING SAMPLES AND TEST SAMPLES FOR EACH CLASS OF LAND COVER IN PAVIA UNIVERSITY

Class No.	Class Color	Class Name	Training	Testing
1		Asphalt	30	6601
2		Meadows	30	18619
3		Gravel	30	2069
4		Trees	30	3034
5		Painted metal sheets	30	1315
6		Bare soil	30	4999
7		Bitumen	30	1300
8		Self-blocking bricks	30	3652
9		Shadows	30	917
		Total	270	42566



(a) (b)

Fig. 6. Visualization of the Pavia University dataset. (a) Pseudocolor map. (b) Ground-truth map.

2) Settings of Baseline Methods: In order to verify the effectiveness of F²HNN, we select several representative baseline methods, including support vector machine (SVM) with RBF kernel, 2-D CNN, 3-D CNN [49], GCN, and FuNet-M [21]. At the same time, two pretrained networks and three networks generated by different feature fusion strategies are also used for comparison experiments.

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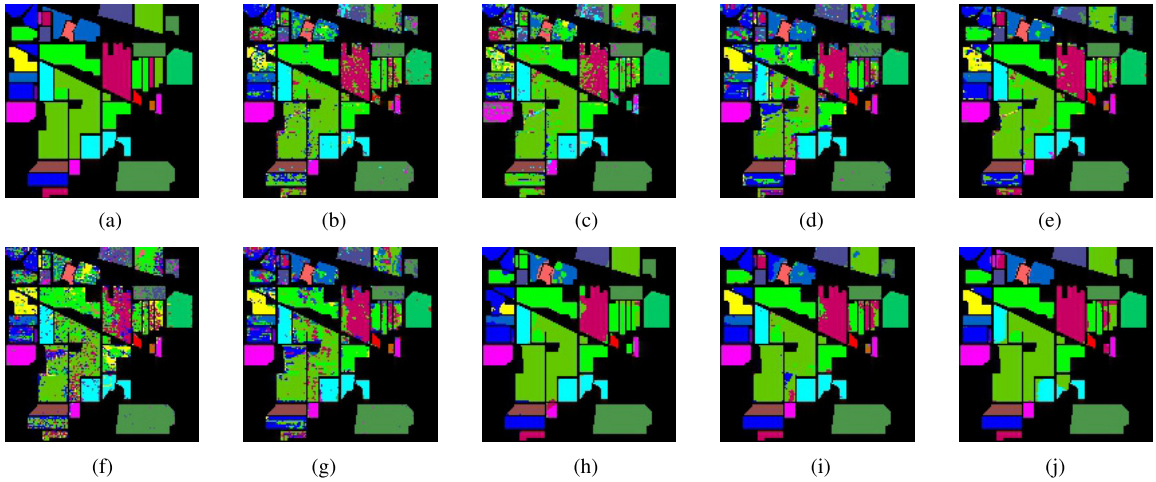


Fig. 7. Visualization of the classification results of different methods on the Indian Pines dataset. (a) gt. (b) SVM. (c) MLP. (d) 2DCNN. (e) 3DCNN. (f) GCN. (g) FuNet-M. (h) F^2HNN_{SS}. (i) F^2HNN_{MS}. (j) F^2HNN_{CSS}.

TABLE V

DETAILED PARAMETER SETTINGS OF THE ENTIRE F^2HNN, INCLUDING THE PRETRAINING NETWORK AND HGNN SETTINGS. THE RELU FUNCTION IS THE ACTIVATION FUNCTION THAT WE USE, FC STANDS FOR THE FULLY CONNECTED LAYER, CONV REPRESENTS THE 3-D CONVOLUTIONAL LAYER, POOL INDICATES THE 3-D POOLING LAYER, BN MEANS THE BATCH NORMALIZATION LAYER, AND HCONV IS THE HYPERGRAPH CONVOLUTIONAL LAYER

		MLP	3D CNN
F^2HNN	Pre-trained model	ReLU	patch_size=7×7
		512×1024 FC	3×3×3 Conv
		ReLU	3×1×1 Pool
		1024×64 FC	3×3×3 Conv
		Softmax	3×1×1 Pool
			3×1×1 Conv
			2×1×1 Conv
			Softmax
		HGNN	
	Follow-up model	Compute $\mathbf{W}$	
		BN	
		C_{in}×128 Hconv	
		BN	
		ReLU	
		128×C_{nClass} Hconv	
		Softmax	

For convenience, we abbreviate the networks generated by the three different strategies as F^2HNN_{SS} (Spectral feature + Spatial feature), F^2HNN_{MS} (MLP feature + Spatial feature), and F^2HNN_{SSC} (Spectral feature + Spatial feature + CNN feature).

C. Classification Results

Tables VI–IX record the overall accuracy (OA), average accuracy (AA), and kappa coefficients of our own method and all baseline methods. Figs. 7–10 are visualizations of all results.

For different datasets, the classification performances of the selected methods are different. Among all baseline methods, 3DCNN achieves the best overall performance. This is due to the powerful feature extraction capabilities of deep learning and the ability of the 3DCNN network to simultaneously learn spatial and spectral modal features. Conventional methods based on SVM and MLP can also obtain an acceptable classification performance, but their performance is not better compared with the methods based on deep learning. As for the baseline method based on GNN, GCN and FuNet-M have completely different performances on different datasets. For the Indian Pines dataset, GCN gets the worst classification accuracy, and FuNet-M's performance is only comparable to that of SVM. However, on the KSC, Botswana, and Pavia University datasets, the performance of GCN exceeds that of SVM and is basically the same as that of 2DCNN, and FuNet-M achieves the highest accuracy of all baseline methods on these three datasets.

A reasonable inference is that the two methods based on GNNs both use spectral features to construct graph structures, that is, pay more attention to the long-distance dependence of HSIs. Comparing the distribution of the labeled samples in the four datasets, it can be found that the samples in the Indian Pines dataset contain more local information, which is contained in the spatial features of HSI and can be well extracted by CNN. In contrast, the distribution of labeled samples in the other three datasets is more discrete, and the long-distance dependence becomes more influential.

Remarkably, the proposed F^2HNN surpassed all baseline methods on the four datasets and achieved the highest OA, AA, and kappa coefficient. On the Indian Pines dataset, F^2HNN can achieve 91.97% OA using only a basic spectral–spatial feature fusion strategy, which is 4.31% higher than the highest one in the baseline methods. Moreover, the classification performance of the three strategies that we designed is incremental. F^2HNN_{MS} uses MLP for preprocessing, which removes the redundancy of spectral information and performs better than F^2HNN_{SS}. F^2HNN_{CSS} fuses the features of the three modalities and introduces CNN features that express local information, achieving the best performance of the three strategies.

D. Hyperparameter Analysis

In this section, we utilize four datasets to quantitatively analyze the hyperparameters k and σ involved in F^2HNN. Theoretically, different k's and σ's can be selected for the

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TABLE VI

PER-CLASS ACCURACY, OA, AA, AND KAPPA COEFFICIENT ACQUIRED BY DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class No.	SVM	MLP	2DCNN	3DCNN [49]	GCN	FuNet-M [21]	F^2HNN_{SS}	F^2HNN_{MS}	F^2HNN_{SSC}
1	48.81	41.23	**100.00**	88.45	91.30	95.65	95.65	**100.00**	97.83
2	72.53	67.52	65.97	87.08	53.64	67.51	**96.85**	93.84	95.59
3	67.15	58.57	71.20	80.03	53.01	63.98	99.40	**99.88**	98.92
4	51.70	49.52	93.24	75.31	87.77	91.98	34.18	98.31	**100.00**
5	86.93	75.18	93.78	90.15	90.89	94.00	87.37	96.27	**100.00**
6	90.65	86.21	89.86	96.34	87.95	92.60	99.04	**99.59**	76.44
7	60.91	45.72	89.28	71.79	85.71	**92.86**	55.79	82.14	81.70
8	94.31	91.73	99.16	97.78	97.07	98.54	**100.00**	**100.00**	**100.00**
9	40.02	49.91	**100.00**	64.31	**100.00**	**100.00**	37.00	**100.00**	50.00
10	71.60	69.88	75.10	86.74	53.81	73.25	79.12	94.34	**96.50**
11	77.73	72.36	61.26	87.12	54.99	61.26	94.46	91.12	**98.33**
12	60.83	45.02	79.25	83.02	38.28	75.72	74.03	71.16	**95.79**
13	95.87	90.50	**100.00**	98.10	98.05	**100.00**	98.54	**100.00**	**100.00**
14	92.74	90.93	91.62	96.83	84.58	94.78	**100.00**	97.79	99.92
15	63.66	53.47	82.21	78.05	65.80	82.90	97.93	98.19	**100.00**
16	91.23	88.55	**100.00**	98.90	97.85	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	77.96	72.13	77.13	87.66	65.97	76.94	91.97	94.17	**96.25**
AA(%)	72.92	67.27	86.99	86.25	77.54	86.56	84.34	**95.16**	93.19
Kappa	0.7475	0.6781	0.7652	0.8603	0.6184	0.7431	0.9003	0.9322	**0.9514**

TABLE VII

PER-CLASS ACCURACY, OA, AA, AND KAPPA COEFFICIENT ACQUIRED BY DIFFERENT METHODS ON THE KENNEDY SPACE CNETER DATASET

Class No.	SVM	MLP	2DCNN	3DCNN [49]	GCN	FuNet-M [21]	F^2HNN_{SS}	F^2HNN_{MS}	F^2HNN_{SSC}
1	90.17	87.91	95.93	95.81	86.33	95.40	**100.00**	**100.00**	99.34
2	86.68	80.81	81.89	83.82	84.36	86.00	94.24	**100.00**	**100.00**
3	69.23	47.32	83.20	78.79	92.19	93.36	**100.00**	98.43	99.61
4	52.96	26.59	48.41	33.00	68.65	55.56	85.32	**100.00**	95.63
5	61.71	55.67	76.40	43.31	75.16	91.93	81.99	81.99	**94.41**
6	40.8	38.94	82.10	68.77	78.17	74.67	**100.00**	97.82	99.13
7	80.81	38.25	**100.00**	89.53	99.04	**100.00**	**100.00**	**100.00**	**100.00**
8	84.53	66.89	77.26	89.74	81.90	87.47	**100.00**	99.30	**100.00**
9	92.87	81.03	98.65	96.91	96.73	99.23	93.27	98.08	**100.00**
10	96.91	89.01	96.78	97.80	90.10	96.53	**100.00**	**100.00**	**100.00**
11	94.79	94.36	99.52	98.71	97.37	98.57	**100.00**	**100.00**	99.76
12	89.84	82.19	93.04	96.48	90.06	90.85	**100.00**	93.44	**100.00**
13	99.93	98.67	**100.00**	**100.00**	99.78	99.89	**100.00**	**100.00**	**100.00**
OA(%)	86.74	77.57	90.75	88.81	89.83	92.43	97.79	98.39	**99.44**
AA(%)	80.14	68.28	87.17	82.51	87.68	89.96	96.52	97.62	**99.07**
Kappa	0.8521	0.7493	0.8738	0.8753	0.8815	0.9058	0.9239	0.9751	**0.9832**

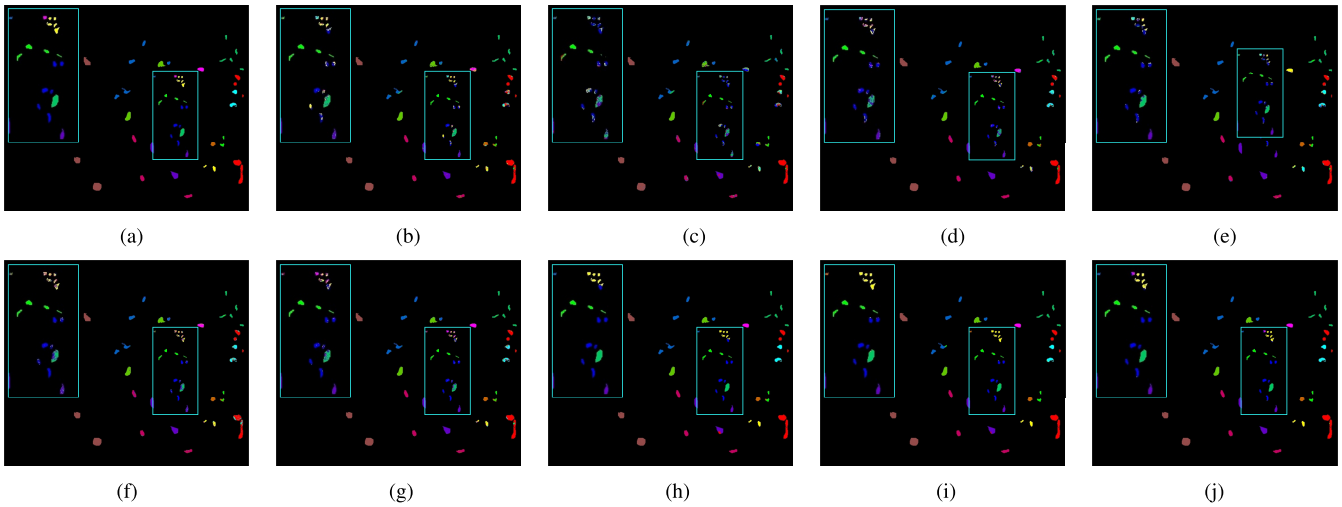


Fig. 8. Visualization of the classification results of different methods on the KSC dataset. (a) gt. (b) SVM. (c) MLP. (d) 2DCNN. (e) 3DCNN. (f) GCN. (g) FuNet-M. (h) F^2HNN_{SS}. (i) F^2HNN_{MS}. (j) F^2HNN_{CSS}.

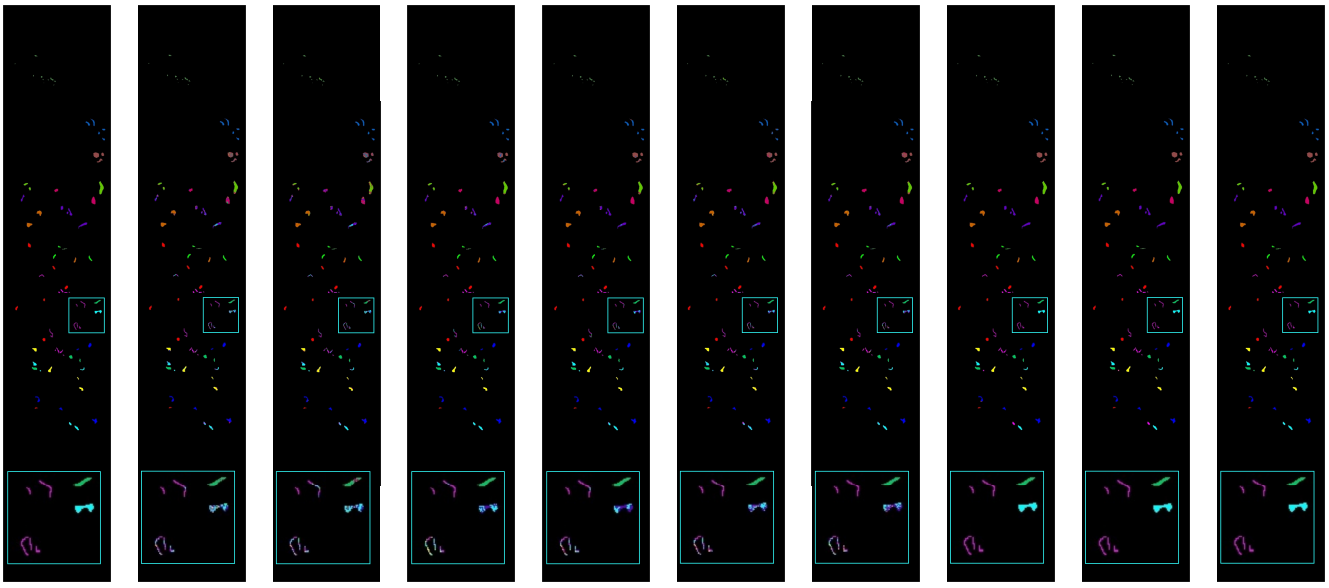
feature of a different modal, but, in order to verify the robustness of the method, k and σ of different modal features should be kept consistent as much as possible under the premise of little impact on the accuracy.

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TABLE VIII

PER-CLASS ACCURACY, OA, AA, AND KAPPA COEFFICIENT ACQUIRED BY DIFFERENT METHODS ON THE BOTSWANA DATASET

Class No.	SVM	MLP	2DCNN	3DCNN [49]	GCN	FuNet-M [21]	F^2HNN_{SS}	F^2HNN_{MS}	F^2HNN_{SSC}
1	99.81	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
2	91.83	97.03	98.02	96.80	98.02	**100.00**	**100.00**	**100.00**	**100.00**
3	91.47	94.02	**100.00**	98.61	98.01	99.60	98.01	**100.00**	**100.00**
4	87.40	88.37	**100.00**	96.95	97.67	99.07	**100.00**	**100.00**	**100.00**
5	82.55	69.52	74.35	87.28	80.67	84.01	**100.00**	**100.00**	**100.00**
6	65.63	75.09	79.93	79.03	65.43	75.09	85.13	85.13	**97.03**
7	97.49	91.12	98.46	99.63	96.14	96.91	**100.00**	**100.00**	**100.00**
8	88.80	92.12	99.01	99.23	98.03	98.52	**100.00**	**100.00**	**100.00**
9	79.64	79.30	82.48	90.10	80.25	87.26	**100.00**	**100.00**	**100.00**
10	79.48	74.60	99.60	93.04	94.76	99.19	**100.00**	**100.00**	**100.00**
11	89.91	73.77	96.07	94.77	86.89	90.49	**100.00**	**100.00**	**100.00**
12	92.44	97.79	98.34	93.86	86.74	98.34	**100.00**	**100.00**	**100.00**
13	84.86	66.42	83.21	95.79	91.42	92.16	**100.00**	**100.00**	**100.00**
14	96.52	86.32	98.95	97.62	82.11	97.89	93.68	**100.00**	93.68
OA(%)	86.81	83.19	92.36	94.02	89.22	93.20	98.43	98.77	**99.57**
AA(%)	85.72	84.68	93.46	94.48	89.72	94.18	98.34	98.94	**99.34**
Kappa	0.8521	0.8173	0.8996	0.9353	0.8745	0.9189	0.9713	0.9804	**0.9951**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Visualization of the classification results of different methods on the Botswana dataset. (a) gt. (b) SVM. (c) MLP. (d) 2DCNN. (e) 3DCNN. (f) GCN. (g) FuNet-M. (h) F^2HNN_{SS}. (i) F^2HNN_{MS}. (j) F^2HNN_{CSS}.

TABLE IX

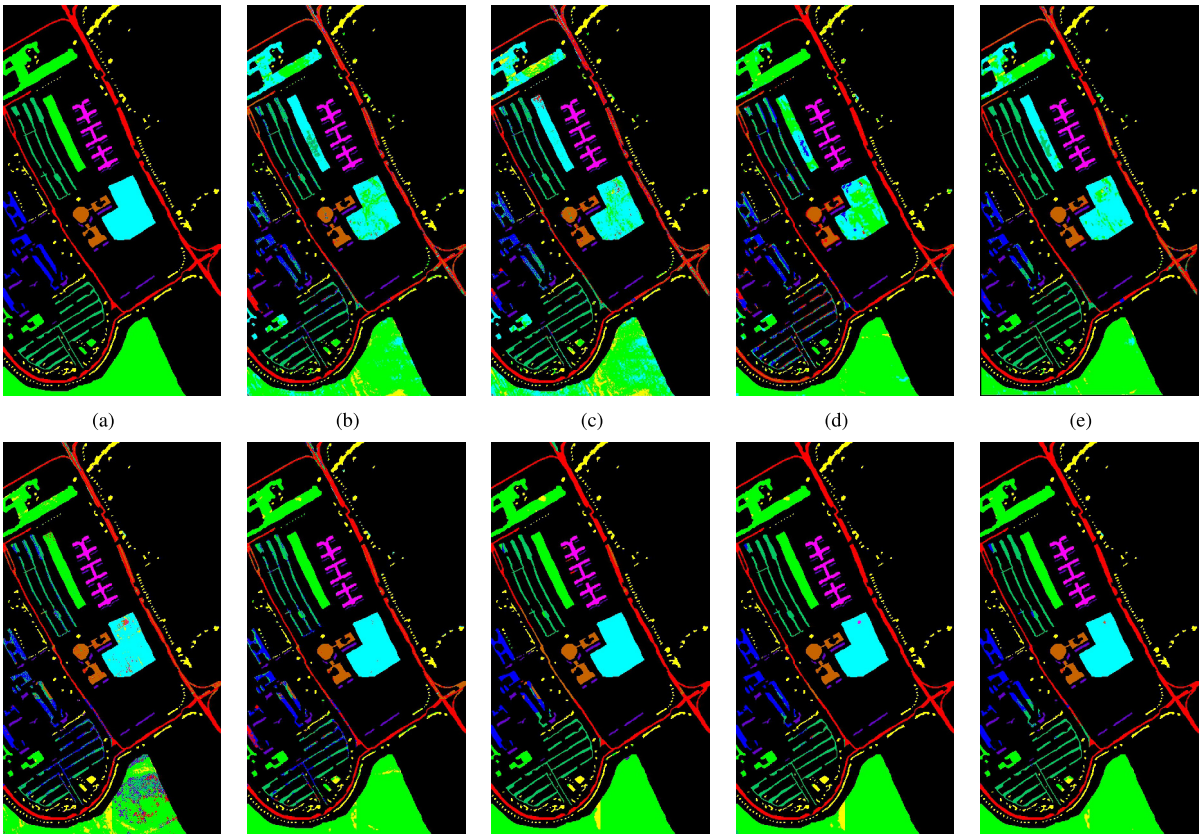
PER-CLASS ACCURACY, OA, AA, AND KAPPA COEFFICIENT ACQUIRED BY DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET

Class No.	SVM	MLP	2DCNN	3DCNN [49]	GCN	FuNet-M [21]	F^2HNN_{SS}	F^2HNN_{MS}	F^2HNN_{SSC}
1	66.26	70.38	86.65	74.56	78.43	81.47	90.35	**98.08**	98.05
2	66.66	52.34	91.25	74.55	77.82	97.49	96.92	96.50	**98.73**
3	56.93	62.79	64.84	77.17	68.13	66.84	85.95	**95.33**	87.47
4	83.65	90.01	90.27	89.26	96.93	91.51	97.68	**99.15**	98.60
5	99.03	97.03	99.93	99.55	99.70	98.66	99.93	99.55	**100.00**
6	76.91	81.39	48.46	87.31	93.96	99.88	**100.00**	99.60	99.74
7	93.76	89.32	72.48	92.63	92.26	96.84	**100.00**	99.70	98.20
8	85.55	90.17	56.52	97.80	67.22	72.79	96.96	**97.39**	97.07
9	99.88	**99.89**	97.25	99.37	**99.89**	99.79	94.09	99.79	98.63
OA(%)	72.77	68.62	80.98	81.13	81.42	91.30	95.91	97.59	**98.06**
AA(%)	80.96	81.48	78.63	88.02	86.04	89.47	95.76	**98.34**	97.39
Kappa	0.6573	0.6169	0.7455	0.7615	0.7655	0.8852	0.9462	0.9683	**0.9744**

1) Analysis of k: Fig. 11 shows the analysis of the hyperparameter k. When analyzing k, σ is fixed to the optimal value shown in Table X. In the experiment, we select the same k value for the features of a different modal. The optimal k value of each dataset is also shown in Table X.

2) Analysis of σ: In our three strategies, there are a total of four different modal features, namely, original spectral feature,

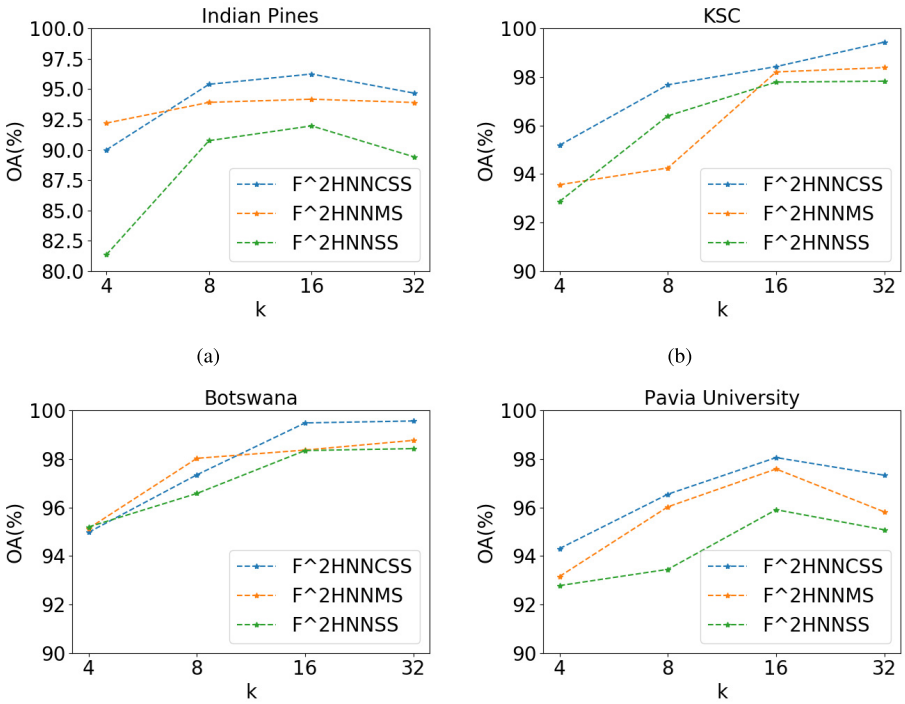
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(a) (b) (c) (d) (e)

(f) (g) (h) (i) (j)

Fig. 10. Visualization of the classification results of different methods on the Pavia University dataset. (a) gt. (b) SVM. (c) MLP. (d) 2DCNN. (e) 3DCNN. (f) GCN. (g) FuNet-M. (h) F^2HNN_{SS}. (i) F^2HNN_{MS}. (j) F^2HNN_{CSS}.



(a) (b) (c) (d)

Fig. 11. Analysis of hyperparameter k on four datasets. (a) Indian Pines. (b) KSC. (c) Botswana. (d) Pavia University.

spatial feature, MLP feature, and CNN feature, and we denote the corresponding σ as σ_1, σ_2, σ_3, and σ_4, respectively.

For the Indian Pines dataset, experiments show that it is sensitive to hyperparameter σ, so we selected different σ's

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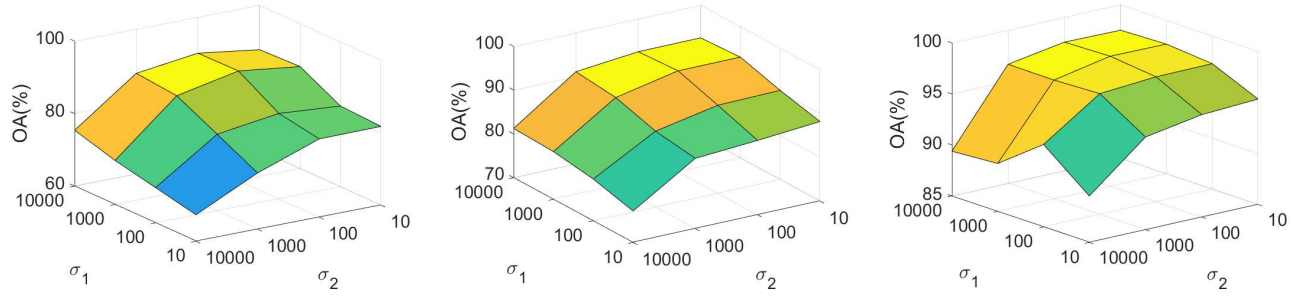


Fig. 12. Analysis of the hyperparameter σ of the three feature fusion strategies on the Indian Pines dataset. σ_1, σ_2, σ_3, and σ_4 represent the σ of spectral features, spatial features, MLP features, and CNN features, respectively. (a) F^2HNN_{SS}. (b) F^2HNN_{MS}. (c) F^2HNN_{CSS}.

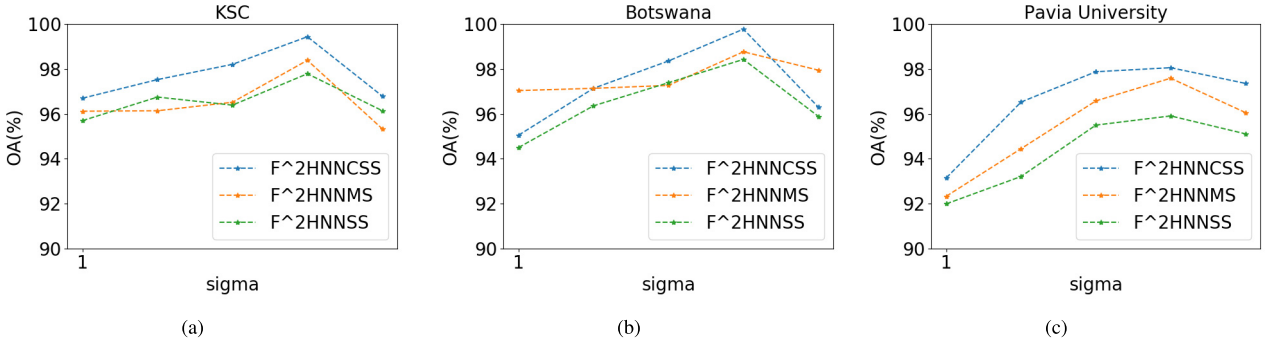


Fig. 13. Analysis of the hyperparameter σ of the three feature fusion strategies on (a) KSC, (b) Botswana, and (c) Pavia University datasets. Here, we set $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4 = \sigma$.

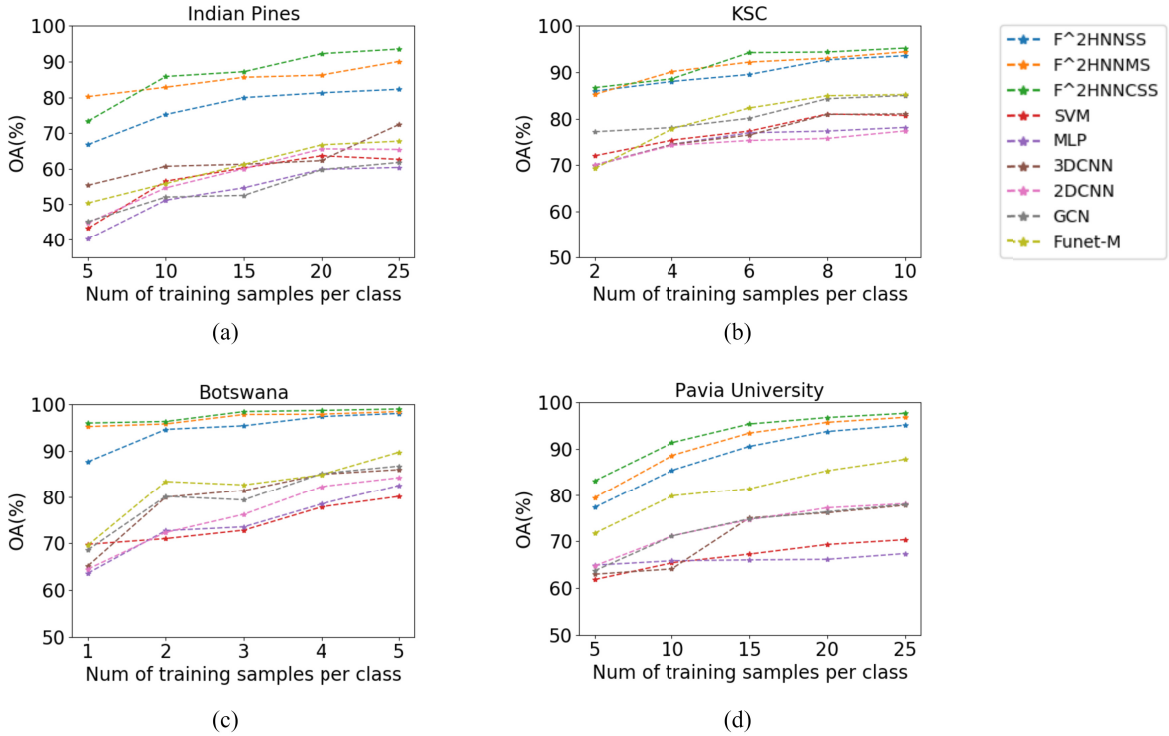


Fig. 14. Analysis of limited training samples classification performance on four datasets. (a) Indian Pines. (b) KSC. (c) Botswana. (d) Pavia University.

for a different modal of features for analysis. Notably, since the impact of σ_1 on classification performance has been fully analyzed in F^2HNN_{SS}, we set σ_1 to a fixed value when analyzing the third strategy F^2HNN_{CSS}. For the KSC, Botswana, and Pavia University datasets, we set $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$ and proved through experiments that it is relatively insensitive to the change of σ. Figs. 10 and 11 show the influence of σ on the four datasets.

By analyzing Figs. 12 and 13, we find that Indian Pines requires a relatively low σ for spatial features compared to

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TABLE X
SELECTION OF k AND σ ON FOUR DATASETS

Name	k	σ_1	σ_2	σ_3	σ_4
Indian Pines	16	1000	100	1000	1000
KSC	32	1000	1000	1000	1000
Botswana	32	1000	1000	1000	1000
Pavia University	16	1000	1000	1000	1000

the other three datasets. The possible reason is that, compared to the other three datasets, Indian Pines has a more compact spatial distribution and more obvious geometric features.

The optimal σ values of different datasets selected for comparison with the baseline methods are shown in Table X.

E. Classification Performance Using Limited Training Samples

In this experiment, we explore the classification performance of different methods with limited training samples. For the Indian Pines and Pavia University datasets, we select five to 25 training samples for each class. For the KSC dataset, the number of training samples is set to two to ten per class. For the Botswana dataset, one to five samples are selected for each class. By analyzing the experimental results in Fig. 14, we can find that the performance of F^{2}HNN far exceeds other compared methods. With extremely limited training samples, the three strategies of F^{2}HNN are still able to achieve considerable classification accuracy.

V. CONCLUSION

In this article, we have proposed a novel F^{2}HNN network for HSI classification. The F^{2}HNN network uses HGNN as a framework and, thus, can dynamically update the hyperedge weights during the network training process. To some extent, this solves the problem of excessive dependence on the construction of the adjacency matrix in the current GCN methods. Furthermore, F^{2}HNN is also very conducive to the fusion of multimodal features, and we then designed three feature fusion strategies for HSI data. Sufficient experiments on four widely used datasets prove that F^{2}HNN surpasses all comparison methods and achieves state-of-the-art performance.

In the future, we plan to design more feature fusion strategies, and we will try to explore the connections and differences between features of a different modal on this basis.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] J. A. Richards and J. Richards, *Remote Sensing Digital Image Analysis*, vol. 3. Berlin, Germany: Springer, 1999.

[3] B. M. Shahshahani and D. A. Landgrebe, "The effect of unlabeled samples in reducing the small sample size problem and mitigating the Hughes phenomenon," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 5, pp. 1087–1095, Sep. 1994.

[4] M. Pal and G. M. Foody, "Feature selection for classification of hyperspectral data by SVM," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 5, pp. 2297–2307, May 2010.

[5] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[6] G. Shaw and D. Manolakis, "Signal processing for hyperspectral image exploitation," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 12–16, Jan. 2002.

[7] J. Xia, L. Bombrun, T. Adali, Y. Berthoumieu, and C. Germain, "Spectral–spatial classification of hyperspectral images using ICA and edge-preserving filter via an ensemble strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4971–4982, Aug. 2016.

[8] Y. Fang *et al.*, "Dimensionality reduction of hyperspectral images based on robust spatial information using locally linear embedding," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1712–1716, Oct. 2014.

[9] X. Kang, S. Li, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[10] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[11] D. Lunga, S. Prasad, M. M. Crawford, and O. Ersoy, "Manifold-learning-based feature extraction for classification of hyperspectral data: A review of advances in manifold learning," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 55–66, Jan. 2014.

[12] H. Yu *et al.*, "Global spatial and local spectral similarity-based manifold learning group sparse representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3043–3056, May 2019.

[13] M. M. Crawford, L. Ma, and W. Kim, "Exploring nonlinear manifold learning for classification of hyperspectral data," in *Optical Remote Sensing*. Berlin, Germany: Springer, 2011, pp. 207–234.

[14] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[15] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[16] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, 2017.

[17] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[18] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 3844–3852.

[19] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[20] F. F. Shahraki and S. Prasad, "Graph convolutional neural networks for hyperspectral data classification," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 968–972.

[21] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 1–13, Jul. 2020.

[22] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[23] Y. Feng, H. You, Z. Zhang, R. Ji, and Y. Gao, "Hypergraph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 3558–3565.

[24] D. Zhou, J. Huang, and B. Schölkopf, "Learning with hypergraphs: Clustering, classification, and embedding," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 19, 2006, pp. 1601–1608.

[25] Q. Fang, J. Sang, C. Xu, and Y. Rui, "Topic-sensitive influencer mining in interest-based social media networks via hypergraph learning," *IEEE Trans. Multimedia*, vol. 16, no. 3, pp. 796–812, Apr. 2014.

[26] D. Li, Z. Xu, S. Li, and X. Sun, "Link prediction in social networks based on hypergraph," in *Proc. 22nd Int. Conf. World Wide Web (WWW)*, 2013, pp. 41–42.

[27] I. Chien, C.-Y. Lin, and I.-H. Wang, "Community detection in hypergraphs: Optimal statistical limit and efficient algorithms," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2018, pp. 871–879.

[28] Y. Wang, L. Zhu, X. Qian, and J. Han, "Joint hypergraph learning for tag-based image retrieval," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4437–4451, Sep. 2018.

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[29] J. Yu, D. Tao, and M. Wang, "Adaptive hypergraph learning and its application in image classification," *IEEE Trans. Image Process.*, vol. 21, no. 7, pp. 3262–3272, Jul. 2012.
[30] L. An, X. Chen, S. Yang, and X. Li, "Person re-identification by multi-hypergraph fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 11, pp. 2763–2774, Nov. 2016.
[31] Z. Zhang, H. Lin, X. Zhao, R. Ji, and Y. Gao, "Inductive multi-hypergraph learning and its application on view-based 3D object classification," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5957–5968, Dec. 2018.
[32] Y. Huang, Q. Liu, and D. Metaxas, "Video object segmentation by hypergraph cut," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1738–1745.
[33] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Jan. 2008.
[34] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016, pp. 1–14.
[35] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 1–12.
[36] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, vol. 2017, pp. 1–19.
[37] J. Jiang, Y. Wei, Y. Feng, J. Cao, and Y. Gao, "Dynamic hypergraph neural networks," in *Proc. IJCAI*, 2019, pp. 2635–2641.
[38] S. Bai, F. Zhang, and P. H. S. Torr, "Hypergraph convolution and hypergraph attention," *Pattern Recognit.*, vol. 110, Feb. 2021, Art. no. 107637.
[39] G. Camps-Valls, T. V. B. Marsheva, and D. Zhou, "Semi-supervised graph-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3044–3054, Oct. 2007.
[40] Y. Gao, R. Ji, P. Cui, Q. Dai, and G. Hua, "Hyperspectral image classification through bilayer graph-based learning," *IEEE Trans. Image Process.*, vol. 23, no. 7, pp. 2769–2778, Jul. 2014.
[41] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, May 2020.
[42] X. He, Y. Chen, and P. Ghamisi, "Dual graph convolutional network for hyperspectral image classification with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 8, 2021, doi: 10.1109/TGRS.2021.3061088.
[43] F. R. Chung and F. C. Graham, *Spectral Graph Theory*, no. 92. Providence, RI, USA: American Mathematical Soc., 1997.
[44] D. Spielman, "Spectral graph theory and its applications," in *Proc. 48th Annu. IEEE Symp. Found. Comput. Sci. (FOCS)*, 2007, pp. 29–38, doi: 10.1109/FOCS.2007.56.
[45] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2020.
[46] R. Hang, Q. Liu, H. Song, and Y. Sun, "Matrix-based discriminant subspace ensemble for hyperspectral image spatial–spectral feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 783–794, Feb. 2015.
[47] M. Liang, L. Jiao, S. Yang, F. Liu, B. Hou, and H. Chen, "Deep multiscale spectral–spatial feature fusion for hyperspectral images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 8, pp. 2911–2924, Aug. 2018.
[48] D. Kingma *et al.*, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015, pp. 1–15.
[49] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.



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